开源软件基础

Python 基础

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Outline

1. 面向对象
2. 函数式特性
   - List Comprehension
   - Map-Reduce
3. 一点更多的数据结构
   - 元组
   - 字典
   - 集合

面向对象

面向对象三要素

- 封装：点号能点出成员
- 继承：儿子点出父亲成员
- 多态：儿子点出和父亲不同的行为

类的声明

```
class ClassName:
    <statement-1>
    .
    .
    .
    <statement-N>
```

__init__ 函数

```
def __init__(self):
    self.data = []
```

Class Objects

```
>>> class Complex:
...     def __init__(self, realpart, imagpart):
...         self.r = realpart
...         self.i = imagpart
...
>>> x = Complex(3.0, -4.5)
>>> x.r, x.i
(3.0, -4.5)
```

成员变量和实例变量

```
class Dog:

    tricks = []             # mistaken use of a class

    def __init__(self, name):
        self.name = name

    def add_trick(self, trick):
        self.tricks.append(trick)

>>> d = Dog('Fido')
>>> e = Dog('Buddy')
>>> d.add_trick('roll over')
```

Class Objects

```
class MyClass:
    """A simple example class"""
    i = 12345

    def f(self):
        return 'hello world'
```

实例/类的成员

- 在 __init__ 中用 self. 初始化的是实例成员
- 在类似 C++ 成员声明处初始化的是类成员（类似静态变量）
- 实例成员可以动态添加和删除
- 必要时用 hasattr 判断是否包含某成员

成员可见性

成员可见性

- 不存在类似 C++ 的私有
- 约定：下划线前缀的是实现细节
- ipython/Jupyter 默认不会补全
- 类似 *nix 下的 dotfiles

继承

```
class A:
    def foo(self):
        print("in A::foo")

class B(A):
    pass

b = B()
b.foo()
```

多态

```python
class A:
    def foo(self):
        print("in A::foo")

class B(A):
    def foo(self):
        print("in B::foo")

b = B()
b.foo()
isinstance(b, B)
isinstance(b, A)
```

鸭子类型

在程序设计中，鸭子类型（英语：duck typing）是动态类型的一种风格。在这种风格中，一个对象有效的语义，不是由继承自特定的类或实现特定的接口，而是由" 当前方法和属性的集合" 决定。这个概念的名字来源于由 James Whitcomb Riley 提出的鸭子测试，“鸭子测试”可以这样表述：

鸭子测试

- If it walks like a duck
- It quacks like a duck
- Then it must be a duck

“当看到一只鸟走起来像鸭子、游泳起来像鸭子、叫起来也像鸭子，那么这只鸟就可以被称为鸭子。”[1]

[1] https://en.wikipedia.org/wiki/Duck_typing

多态

```
class A:
    def foo(self):
        print("in A::foo")

class B(A):
    def foo(self):
        print("in B::foo")

class C:
    def foo(self):
        print("not subclass of A/B")

def test_class(a):
    a.foo()
```

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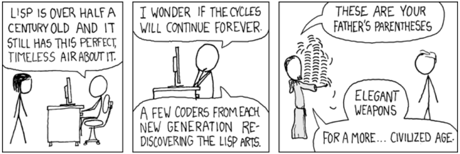
函数一等公民

- 可以被存入变数或其他结构
- 可以被作为参数传递给其他函数
- 可以被作为函数的返回值
- 可以在执行期创造，而无需完全在设计期全部写出
- 即使没有被系结至某一名称，也可以存在

Lisp

```
(+ 1 2)
(define (foo bar) (+ bar 1))
(foo 100)
(define (fib x)
    (if (<= x 1)
        1
        (+ (fib (- x 1))
             (fib (- x 2)))))
(display (fib 10))
```



Haskell

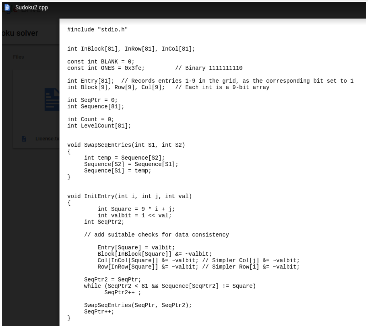
- 偏学术
- 号称最纯粹的函数式语言
- Avoid Success at all Cost

今天就能开始用的小知识

关于 Haskell

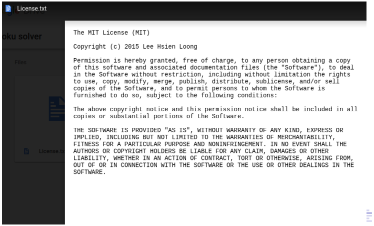
- Avoid Success at all Cost, or Avoid "Success at all cost"
- 新加坡总理李显龙（会 C/C++）称退休后会学习 Haskell [a]

[a] http://www.pmo.gov.sg/newsroom/transcript-speech-prime-minister-lee-hsien-loong-founders-forum-s



Sudoku2.cpp

```
#include "stdio.h"


int InBlock[81], InRow[81], InCol[81];

const int BLANK = 0;
const int ONES = 0x3fe;        // Binary 1111111110

int Entry[81];  // Records entries 1-9 in the grid, as the corresponding bit set to 1
int Block[9], Row[9], Col[9];   // Each int is a 9-bit array

int SeqPtr = 0;
int Sequence[81];

int Count = 0;
int LevelCount[81];


void SwapSeqEntries(int S1, int S2)
{
     int temp = Sequence[S2];
     Sequence[S2] = Sequence[S1];
     Sequence[S1] = temp;
}


void InitEntry(int i, int j, int val)
{
         int Square = 9 * i + j;
         int valbit = 1 << val;
     int SeqPtr2;

     // add suitable checks for data consistency

         Entry[Square] = valbit;
         Block[InBlock[Square]] &= ~valbit;
         Col[InCol[Square]] &= ~valbit; // Simpler Col[j] &= ~valbit;
         Row[InRow[Square]] &= ~valbit; // Simpler Row[i] &= ~valbit;

     SeqPtr2 = SeqPtr;
     while (SeqPtr2 < 81 && Sequence[SeqPtr2] != Square)
           SeqPtr2++ ;

     SwapSeqEntries(SeqPtr, SeqPtr2);
     SeqPtr++;
}
```



License.txt

```
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```

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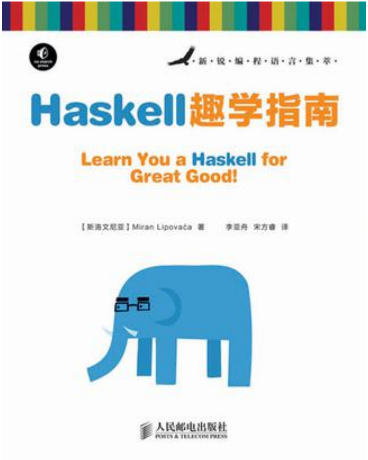
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什么是函数式

函数式编程（functional programming）或称函数程序设计，是一种编程典范，它将电脑运算视为数学上的函数计算，并且避免使用程序状态以及易变对象。函数编程语言最重要的基础是 λ 演算（lambda calculus）。而且 λ 演算的函数可以接受函数当作输入（引数）和输出（传出值）。
比起命令式编程，函数式编程更加强调程序执行的结果而非执行的过程，倡导利用若干简单的执行单元让计算结果不断渐进，逐层推导复杂的运算，而不是设计一个复杂的执行过程。

函数式特性

- First-class and higher-order functions
- Pure function
- Recursion
- Type system
- ...

λ 演算

λ 演算（英语：lambda calculus，λ-calculus）是一套从数学逻辑中发展，以变量绑定和替换的规则，来研究函数如何抽象化定义、函数如何被应用以及递归的形式系统。它由数学家阿隆佐·邱奇在 20 世纪 30 年代首次发表。Lambda 演算作为一种广泛用途的计算模型，可以清晰地定义什么是一个可计算函数，而任何可计算函数都能以这种形式表达和求值，它能模拟单一磁带图灵机的计算过程；尽管如此，Lambda 演算强调的是变换规则的运用，而非实现它们的具体机器。

λ 表达式

```
inc = lambda x: x + 1

add = lambda x, y: x + y

inc(5)
add(10, 20)

# 注意：Python的lambda表达式不能换行
```

Outline

列表推导

```
[i * 2 for i in range(10)]

[i * 2 for i in range(10) if i % 2 == 0]

colours = ["red", "green", "yellow", "blue"]
things = ["house", "car", "tree"]
combined =  [(x,y) for x in colours for y in things]
```

Outline

Map/Filter/Reduce

```
map(lambda x: x ** 2, range(10)
list(map(lambda x: x **2, range(10))

filter(lambda x: x % 2 == 0,
        map(lambda x: x ** 2, range(10)))

reduce(lambda x, y: x + y,
        filter(lambda x: x % 2 == 0,
                map(lambda x: x ** 2, range(10))))
```

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Tuples and Sequences

- 元组与列表类似，都是线性数据结构
- 元组由逗号分割的数据构成
- 元组内容不可变
- 元组可以嵌套

Tuples and Sequences

```
>>> t = 12345, 54321, 'hello!'
>>> t[0]
12345
>>> t
(12345, 54321, 'hello!')
>>> # Tuples may be nested:
... u = t, (1, 2, 3, 4, 5)
>>> u
((12345, 54321, 'hello!'), (1, 2, 3, 4, 5))
>>> # Tuples are immutable:
... t[0] = 88888
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'tuple' object does not support item assignment
>>> # but they can contain mutable objects:
... v = ([1, 2, 3], [3, 2, 1])
>>> v
([1, 2, 3], [3, 2, 1])
```

Tuples and Sequences

- 构造空列表使用 ()
- 构造单元素列表用 (a,)
- 如果元组元素是可变的，则可以修改（有坑）
- 元组也可出现在赋值左部（lhs）

Tuples and Sequences

```
>>> empty = ()
>>> singleton = 'hello',    # <-- note trailing comma
>>> len(empty)
0
>>> len(singleton)
1
>>> singleton
('hello',)
```

Tuples and Sequences

```
>>> x, y, z = t
>>> x, y = y, x

>>> x = (1, 2, 3, [4, 5])
>>> x[3] += [6]
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
<ipython-input-34-1b7fadc4e686> in <module>()
----> 1 x[3] += [6]

TypeError: 'tuple' object does not support item assignment

>>> x
(1, 2, 3, [4, 5, 6])
```

Outline

Dictionaries

- 字典类似其他语言中散列表、关联数组，按键值方式组织
- 不像列表元组可以用下标随机访问
- 任何不可变对象均可作为键（元组亦可，但不能包含可变元素）
- 空字典用 {} 构造；非空字典在 {} 内逗号分割的键: 值对表示
- list(d.keys()) 返回所有键。如需有序，使用 sorted(d.keys())

Dictionaries

```
>>> tel = {'jack': 4098, 'sape': 4139}
>>> tel['guido'] = 4127
>>> tel
{'sape': 4139, 'guido': 4127, 'jack': 4098}
>>> tel['jack']
4098
>>> del tel['sape']
>>> tel['irv'] = 4127
>>> tel
{'guido': 4127, 'irv': 4127, 'jack': 4098}
>>> list(tel.keys())
['irv', 'guido', 'jack']
>>> sorted(tel.keys())
['guido', 'irv', 'jack']
>>> 'guido' in tel
True
>>> 'jack' not in tel
False
```

Dictionaries

```
>>> x = {}
>>> x[(1, 2, 3)] = 4
>>> x[(1, 2, [3, 4])] = 5
>>> x[(1, 2, 3, [4, 5])] = 6
---------------------------------------------------------------------------
TypeError         Traceback (most recent call last)
<ipython-input-8-7e134042fef1> in <module>()
----> 1 x[(1, 2, 3, [4, 5])] = 6

TypeError: unhashable type: 'list'
```

Dictionaries

- 字典还可以由二元组列表初始化
- 字典推导与列表推导类似
- 字典键为字符串时，可以用关键字参数设定

Dictionaries

```
>>> dict([('sape', 4139),
          ('guido', 4127),
          ('jack', 4098)])
{'sape': 4139, 'jack': 4098, 'guido': 4127}
```

Dictionaries

```
>>> {x: x**2 for x in (2, 4, 6)}
{2: 4, 4: 16, 6: 36}
```

Dictionaries

```
>>> dict(sape=4139, guido=4127, jack=4098)
{'sape': 4139, 'jack': 4098, 'guido': 4127}
```

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Sets

- 集合是无序无重复的数据结构
- 支持成员测试、添加、删除等操作
- 支持交集、并集、差等集合操作
- 初始化用 {} 括住的逗号分割元素序列
- 空集合用 set()，而不是 {}
- 集合推导同样支持

Sets

```
>>> basket = {'apple', 'orange', 'apple', 'pear', 'orange', 'banana'}
>>> print(basket)                      # show that duplicates have been re
{'orange', 'banana', 'pear', 'apple'}
>>> 'orange' in basket                 # fast membership testing
True
>>> 'crabgrass' in basket
False

>>> # Demonstrate set operations on unique letters from two words
...
>>> a = set('abracadabra')
>>> b = set('alacazam')
>>> a                                  # unique letters in a
{'a', 'r', 'b', 'c', 'd'}
>>> a - b                              # letters in a but not in b
{'r', 'd', 'b'}
>>> a | b                              # letters in a or b or both
{'a', 'c', 'r', 'd', 'b', 'm', 'z', 'l'}
>>> a & b                              # letters in both a and b
{'a', 'c'}
>>> a ^ b                              # letters in a or b but not both
{'r', 'd', 'b', 'm', 'z', 'l'}
```

Sets

```
>>> a = {x for x in 'abracadabra' if x not in 'abc'}
>>> a
{'r', 'd'}
```